

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
Student Name:	PRITHAM G	Reg. No.:	192572014
Section / Batch:	CSE -AI	Date:	24/07/2026

Code of Conduct

I PRITHAM G (192572014) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: Pritham g

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25

Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marks**Q2**
Stack
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Stack - Balancing Symbols
5 marks**Q6**
Singly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Linked List
5 marks**Q9**
Linked List
5 marks**Q10**
Queue
5 marks

10/10 answered

Q1 Binary Search

5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {  
    int low = 0, high = n - 1;  
    while(low < high) {  
        int mid = (low + high) / 2;  
        if(arr[mid] == key)  
            return mid;  
        else if(arr[mid] < key)  
            low = mid;  
        else  
            high = mid;  
    }  
    return -1;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int binarySearch(int arr[], int n, int key) {  
4     int low = 0, high = n - 1;  
5     while (low <= high) {  
6         int mid = (low + high) / 2;  
7         if (arr[mid] == key)  
8             return mid;  
9         else if (arr[mid] < key)  
10            low = mid + 1;  
11        else  
12            high = mid - 1;  
13    }  
14    return -1;  
15 }  
16  
17 int main() {  
18     int arr[] = {2, 4, 6, 8, 10, 12, 14, 16, 18, 20};  
19     int n = sizeof(arr) / sizeof(arr[0]);  
20  
21     int testKeys[] = {10, 2, 20, 7, 18, 1, 21};  
22     int numTests = sizeof(testKeys) / sizeof(testKeys[0]);  
23  
24     for (int i = 0; i < numTests; i++) {  
25         int key = testKeys[i];  
26         int result = binarySearch(arr, n, key);  
27  
28         if (result != -1)  
29             printf("Key %d found at index %d\n", key, result);  
30         else  
31             printf("Key %d not found in array\n", key);  
32     }
```

Test Input

stdin input (optional)

Output

```
Key 10 found at index 4  
Key 2 found at index 0  
Key 20 found at index 9  
Key 7 not found in array  
Key 18 found at index 8  
Key 1 not found in array
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {  
    if(top == 0) {  
        printf("Underflow");  
        return -1;  
    }  
    return stack[top--];  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int stack[MAX];  
6 int top = -1;  
7  
8 void push(int value) {  
9     if(top == MAX - 1) {  
10        printf("Overflow\n");  
11        return;  
12    }  
13    stack[++top] = value;  
14 }  
15  
16 int pop() {  
17     if(top == -1) {  
18        printf("Underflow\n");  
19        return -1;  
20    }  
21    return stack[top--];  
22 }  
23  
24 int main() {  
25  
26     push(10);  
27     push(20);  
28     push(30);  
29  
30     printf("Popped: %d\n", pop());  
31     printf("Popped: %d\n", pop());  
32     printf("Popped: %d\n", pop());  
33  
34  
35     printf("Popped: %d\n", pop());  
36     return 0;  
37 }
```

Test Input

stdin input (optional)

Output

```
Popped: 30  
Popped: 20  
Popped: 10  
Underflow  
Popped: -1
```

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🟢 Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[5];
4 int top;
5
6 void init() {
7     top = -1;
8 }
9
10 int push(int x) {
11     if (top == 4) {
12         return 0;
13     }
14     stack[++top] = x;
15     return 1;
16 }
17
18 int pop(int *val) {
19     if (top == -1) {
20         return 0;
21     }
22     *val = stack[top--];
23     return 1;
24 }
25
26 int peek(int *val) {
27     if (top == -1) return 0;
28     *val = stack[top];
29     return 1;
30 }
31
32 void printStack() {
33     printf("Stack (top=%d): ", top);
34     for (int i = 0; i <= top; i++) {
35         printf("%d ", stack[i]);
36     }
37     printf("\n");
38 }
39
40 int main() {
41     int val;
42 }
```

Test Input

stdin input (optional)

Output

```
---- Test Case 1: Basic
Push/Pop ----
Stack (top=2): 10 20 30
Popped: 30
Popped: 20
Stack (top=0): 10
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stacks
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q4 Stack

5 marks

Identify the errors in the following snippet.

```
if(ch == ' '){  
while(stack[top] != ' '){  
postfix[j++] = pop();  
}  
pop();  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <ctype.h>  
3 #include <string.h>  
4  
5 #define MAX 100  
6  
7 char stackArr[MAX];  
8 int top = -1;  
9  
10 void push(char c) {  
11     if (top < MAX - 1) stackArr[++top] = c;  
12 }  
13  
14 char popChar() {  
15     if (top >= 0) return stackArr[top--];  
16     return '\0';  
17 }  
18  
19 int isEmpty() {  
20     return top == -1;  
21 }  
22  
23 int precedence(char op) {  
24     if (op == '^') return 3;  
25     if (op == '*' || op == '/') return 2;  
26     if (op == '+' || op == '-') return 1;  
27     return 0;  
28 }  
29  
30 void infixToPostfix(const char *infix, char *postfix) {  
31     int i = 0, j = 0;  
32  
33     while (infix[i] != '\0') {  
34         char ch = infix[i];  
35  
36         if (isalnum(ch)) {  
37             postfix[j++] = ch;  
38         }  
39         else if (ch == '(') {  
40             push(ch);  
41         }
```

Test Input

(A+B)*C

Output

Enter infix expression (no spaces):
Postfix: AB+C*

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q5 Stack - Balancing Symbols

5 marks

In balancing Bracket, you identify the error.

```
if((ch == '[' && stack[top] == '[') ||  
(ch == ']' && stack[top] == '[')) {  
    pop();  
} else {  
    return 0;  
}
```

🔍 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <string.h>  
3 #include <stdbool.h>  
4  
5  
6 int isBalanced(char* expr) {  
7     int n = strlen(expr);  
8     char stack[n];  
9     int top = -1;  
10  
11     for (int i = 0; i < n; i++) {  
12         char ch = expr[i];  
13  
14  
15         if (ch == '(' || ch == '[' || ch == '{') {  
16             stack[++top] = ch;  
17         }  
18  
19         else if (ch == ')' || ch == ']' || ch == '}') {  
20  
21             if (top == -1) {  
22                 return 0;  
23             }  
24  
25  
26             if ((ch == ')') && stack[top] == '(') ||  
27                 (ch == ']') && stack[top] == '[') ||  
28                 (ch == '}') && stack[top] == '{') {  
29                 top--;  
30             } else {  
31                 return 0;  
32             }  
33         }  
34     }  
35  
36     return (top == -1);  
37 }  
38  
39  
40 int main() {  
41
```

Test Input

stdin input (optional)

Output

```
--- Bracket Balancing Test  
Results ---  
Input: {([)]} | Result:  
Balanced  
Input: {([)]} | Result: Not  
Balanced
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q6 Singly Linked List

5 marks

What is the issue?

temp = head

while temp.next != NULL:

print(temp.data)

temp = temp.next

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 typedef struct Node {
5     int data;
6     struct Node* next;
7 } Node;
8
9 Node* createNode(int data) {
10     Node* newNode = (Node*)malloc(sizeof(Node));
11     newNode->data = data;
12     newNode->next = NULL;
13     return newNode;
14 }
15
16 void printList(Node* head) {
17     Node* temp = head;
18
19     while (temp != NULL) {
20         printf("%d ", temp->data);
21         temp = temp->next;
22     }
23     printf("\n");
24 }
25
26
27 int main() {
28
29     Node* head = createNode(1);
30     head->next = createNode(2);
31     head->next->next = createNode(3);
32     head->next->next->next = createNode(4);
33
34     printf("Output:\n");
35     printList(head);
36
37     return 0;
38 }
```

Test Input

stdin input (optional)

Output

Output:
1 2 3 4

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Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q7 Stack

5 marks

Point out the error.

```
POP(stack):  
if top == 0:  
    print("Underflow")  
else:  
    return stack[top--]
```

🔍 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int stack[MAX];  
6 int top = -1;  
7  
8 int push(int x) {  
9     if (top == MAX - 1) {  
10         return 0;  
11     }  
12     stack[++top] = x;  
13     return 1;  
14 }  
15  
16 int pop(int *out) {  
17     if (top == -1) {  
18         return 0;  
19     }  
20     *out = stack[top];  
21     top--;  
22     return 1;  
23 }  
24  
25 int main() {  
26     int val;  
27  
28     printf("Test 1: Pop from empty stack\n");  
29     if (!pop(&val)) printf("Underflow\n");  
30     else printf("Popped: %d\n", val);  
31  
32     printf("\nTest 2: Push 10, 20 then pop\n");  
33     push(10);  
34     push(20);  
35  
36     if (pop(&val)) printf("Popped: %d\n", val);  
37     if (pop(&val)) printf("Popped: %d\n", val);  
38  
39     printf("\nTest 3: Pop after stack becomes empty\n");
```

Test Input

stdin input (optional)

Output

```
Test 1: Pop from empty stack  
Underflow  
  
Test 2: Push 10, 20 then pop  
Popped: 20  
Popped: 10
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

[← Back](#)

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q8 Linked List

5 marks

Identify the errors in the following code

while current != NULL:

current.next = prev

prev = current

current = current.next

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4
5 struct Node {
6     int data;
7     struct Node* next;
8 };
9
10
11 void reverseList(struct Node** head_ref) {
12     struct Node* prev = NULL;
13     struct Node* current = *head_ref;
14     struct Node* next_node = NULL;
15
16
17     while (current != NULL) {
18         next_node = current->next;
19         current->next = prev;
20         prev = current;
21         current = next_node;
22     }
23     *head_ref = prev;
24 }
25
26
27 void printList(struct Node* node) {
28     while (node != NULL) {
29         printf("%d -> ", node->data);
30         node = node->next;
31     }
32     printf("NULL\n");
33 }
34
35
36 void push(struct Node** head_ref, int new_data) {
37     struct Node* new_node = (struct Node*)malloc(sizeof(struct Node));
38     new_node->data = new_data;
39     new_node->next = (*head_ref);
40     (*head_ref) = new_node;
41 }
```

Test Input

stdin input (optional)

Output

Original Linked List:
40 -> 30 -> 20 -> 10 -> NULL

Reversed Linked List:
10 -> 20 -> 30 -> 40 -> NULL

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q9 Linked List

5 marks

What is the error in the below code?

```
new.next = head  
head = new
```

🔍 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3  
4 struct Node {  
5     int data;  
6     struct Node *next;  
7 };  
8  
9 void insertAtBeginning(struct Node **head, int value) {  
10     struct Node *newNode = (struct Node *)malloc(sizeof(struct Node));  
11     if (newNode == NULL) {  
12         printf("Memory allocation failed\n");  
13         return;  
14     }  
15     newNode->data = value;  
16     newNode->next = *head;  
17     *head = newNode;  
18 }  
19  
20 void printList(struct Node *head) {  
21     while (head != NULL) {  
22         printf("%d ", head->data);  
23         head = head->next;  
24     }  
25     printf("\n");  
26 }  
27  
28 int main() {  
29     struct Node *head = NULL;  
30  
31  
32     insertAtBeginning(&head, 10);  
33     insertAtBeginning(&head, 20);  
34     insertAtBeginning(&head, 30);  
35  
36     printf("Original List: ");  
37     printList(head);  
38  
39  
40     insertAtBeginning(&head, 5);  
41  
42     printf("After inserting 5 at beginning: ");  
43     printList(head);  
44  
45     return 0;  
46 }
```

Test Input

stdin input (optional)

Output

```
Original List: 30 20 10  
After inserting 5 at beginning: 5  
30 20 10
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q10 Queue

5 marks

What is missing in the below pseudocode?

```
DEQUEUE(Q):  
if front == -1:  
    print("Underflow")  
else:  
    return Q[front++]
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #define MAX 5  
3  
4 struct Queue {  
5     int items[MAX];  
6     int front;  
7     int rear;  
8 };  
9  
10  
11 void initQueue(struct Queue* q) {  
12     q->front = -1;  
13     q->rear = -1;  
14 }  
15  
16  
17 int isFull(struct Queue* q) {  
18     return q->rear == MAX - 1;  
19 }  
20  
21  
22 int isEmpty(struct Queue* q) {  
23     return q->front == -1 || q->front > q->rear;  
24 }  
25  
26  
27 void enqueue(struct Queue* q, int value) {  
28     if (isFull(q)) {  
29         printf("Overflow\n");  
30     } else {  
31         if (q->front == -1) q->front = 0;  
32         q->rear++;  
33         q->items[q->rear] = value;  
34         printf("Enqueued: %d\n", value);  
35     }  
36 }  
37  
38 int dequeue(struct Queue* q) {  
39     int item;  
40     if (q->front == -1 || q->front > q->rear) {  
41         printf("Underflow\n");  
42         return -1;  
43     }  
44 }
```

Test Input

stdin input (optional)

Output

```
--- Test Case: Standard  
Operations ---  
Enqueued: 10  
Enqueued: 20  
Enqueued: 30  
Enqueued: 40
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks

10/10 answered

Q1 Stack

5 marks

Point out the error?

PEEK(stack):
return stack[top+1]

🔍 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAX 100
5
6 int stack[MAX];
7 int top = -1;
8
9
10 void push(int value) {
11     if (top == MAX - 1) {
12         printf("Stack Overflow\n");
13         return;
14     }
15     stack[++top] = value;
16 }
17
18
19 int pop() {
20     if (top == -1) {
21         printf("Stack Underflow\n");
22         return -1;
23     }
24     return stack[top--];
25 }
26
27
28 int peek() {
29     if (top == -1) {
30         printf("Stack is empty\n");
31         return -1;
32     }
33     return stack[top];
34 }
35
36 int main() {
37     push(10);
38     push(20);
39     push(30);
40
41     printf("Top element is: %d\n", peek());
```

Test Input

stdin input (optional)

Output

Top element is: 30
Top element after pop is: 20

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks

10/10 answered

Q2 Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
    printf("Queue Full");
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queueArr[MAX];
6 int front = 0, rear = -1;
7
8
9 int isFull() {
10     return ((rear + 1) % MAX == front);
11 }
12
13
14 int isEmpty() {
15     return (rear == -1);
16 }
17
18
19 void enqueue(int vehicle) {
20     if (isFull()) {
21         printf("Queue Full\n");
22         return;
23     }
24     if (rear == -1) {
25         front = 0;
26         rear = 0;
27     } else {
28         rear = (rear + 1) % MAX;
29     }
30     queueArr[rear] = vehicle;
31     printf("Enqueued: %d\n", vehicle);
32 }
33
34 void display() {
35     if (isEmpty()) {
36         printf("Queue Empty\n");
37         return;
38     }
39 }
```

Test Input

stdin input (optional)

Output

```
TEST CASE 1: Enqueue 6 items
into size 5 circular queue
Queue Full
Queue Empty
Queue Full
Queue Empty
```

🔴 Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marks**Q2**
Circular Queue
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Doubly Linked List
5 marks**Q6**
Queue
5 marks**Q7**
Queue - Deque
5 marks**Q8**
Binary Search
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack - Postfix Expression
5 marks

10/10 answered

Q3 Stack

5 marks

Point out the error.
if top == MAX:
Overflow

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #define MAX 5
3
4 int stack_arr[MAX];
5 int top = -1;
6
7
8 void push(int x) {
9
10     if (top == MAX - 1) {
11         printf("Overflow! Cannot push %d\n", x);
12         return;
13     }
14     stack_arr[++top] = x;
15     printf("Pushed %d\n", x);
16 }
17
18
19 void pop() {
20     if (top == -1) {
21         printf("Underflow! Stack is empty\n");
22         return;
23     }
24     int item = stack_arr[top--];
25     printf("Popped %d\n", item);
26 }
27
28
29 void display() {
30     if (top == -1) {
31         printf("Stack is empty\n");
32         return;
33     }
34     printf("Stack elements: ");
35     for (int i = 0; i <= top; i++) {
36         printf("%d ", stack_arr[i]);
37     }
38     printf("\n");
39 }
40
41 int main() {
42
43     printf("=== Test Case: Overflow check ===\n");
44     push(10);
45     push(20);
46     push(30);
```

Test Input

stdin input (optional)

Output

```
=== Test Case: Overflow check ===
Pushed 10
Pushed 20
Pushed 30
Pushed 40
```

⚠ Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marks**Q2**
Circular Queue
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Doubly Linked List
5 marks**Q6**
Queue
5 marks**Q7**
Queue - Deque
5 marks**Q8**
Binary Search
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack - Postfix Expression
5 marks

10/10 answered

Q4 Stack

5 marks

Identify the issue in the below pseudocode.

POP():

if top == 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stackArr[MAX];
6 int top = -1;
7
8 int pop() {
9     if (top == -1) {
10
11         printf("Underflow: stack is empty\n");
12         return -1;
13     }
14     int x = stackArr[top];
15     top--;
16     return x;
17 }
18
19 void push(int val) {
20     if (top == MAX - 1) {
21         printf("Overflow: stack is full\n");
22         return;
23     }
24     stackArr[++top] = val;
25 }
26
27 int main() {
28
29     printf("Test 1: POP on empty stack\n");
30     int x1 = pop();
31     printf("Returned: %d\n", x1);
32
33
34     printf("Test 2: Push 10, 20, 30 then POP three times\n");
35     push(10);
36     push(20);
37     push(30);
38
39     printf("POP: %d\n", pop());
40     printf("POP: %d\n", pop());
41     printf("POP: %d\n", pop());
42     printf("\n");
43
44
45     printf("Test 3: POP after stack becomes empty\n");
46     int x3 = pop();
```

Test Input

stdin input (optional)

Output

```
Test 1: POP on empty stack
Underflow: stack is empty
Returned: -1

Test 2: Push 10, 20, 30 then
POP three times
```

Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Postfix Expression
5 marks

10/10 answered

Q5 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {  
    if (del->prev != NULL)  
        del->prev->next = del->next;  
    if (del->next != NULL)  
        del->next->prev = del->prev;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3  
4 typedef struct Node {  
5     int data;  
6     struct Node* prev;  
7     struct Node* next;  
8 } Node;  
9  
10 Node* createNode(int data) {  
11     Node* n = (Node*)malloc(sizeof(Node));  
12     n->data = data;  
13     n->prev = NULL;  
14     n->next = NULL;  
15     return n;  
16 }  
17  
18 void append(Node** head, int data) {  
19     Node* newNode = createNode(data);  
20     if (*head == NULL) {  
21         *head = newNode;  
22         return;  
23     }  
24     Node* cur = *head;  
25     while (cur->next != NULL) cur = cur->next;  
26     cur->next = newNode;  
27     newNode->prev = cur;  
28 }  
29  
30 void printList(Node* head) {  
31     Node* cur = head;  
32     while (cur != NULL) {  
33         printf("%d ", cur->data);  
34         cur = cur->next;  
35     }  
36     printf("\n");  
37 }  
38  
39 void deleteNode(Node** headRef, Node* del) {  
40     if (del == NULL) return;  
41 }
```

Test Input

stdin input (optional)

Output

```
Original list: 1 2 3 4 5  
After deleting 3: 1 2 4 5  
After deleting 1 (head): 2 4 5  
After deleting 5 (tail): 2 4
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Postfix Expression
5 marks

10/10 answered

Q6 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {  
    if (rear == MAX) return;  
    queue[rear++] = data;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int queueArr[MAX];  
6 int front = 0;  
7 int rear = -1;  
8  
9 void enqueue(int data) {  
10  
11     if (rear == MAX - 1) return;  
12     queueArr[++rear] = data;  
13 }  
14  
15 int dequeue(void) {  
16     if (front > rear) return -1;  
17     return queueArr[front++];  
18 }  
19  
20 void display(void) {  
21     if (front > rear) {  
22         printf("Queue: Empty\n");  
23         return;  
24     }  
25     printf("Queue: ");  
26     for (int i = front; i <= rear; i++) {  
27         printf("%d ", queueArr[i]);  
28     }  
29     printf("\n");  
30 }  
31  
32 int main() {  
33  
34     printf("Test 1: Enqueue until full\n");  
35     enqueue(10);  
36     enqueue(20);  
37     enqueue(30);  
38     enqueue(40);  
39     enqueue(50);  
40  
41  
42     display();  
43  
44 }
```

Test Input

stdin input (optional)

Output

Test 1: Enqueue until full
Queue: 10 20 30 40 50Test 2: Dequeue two, then
enqueue two
Dequeued: 10

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marks**Q2**
Circular Queue
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Doubly Linked List
5 marks**Q6**
Queue
5 marks**Q7**
Queue - Deque
5 marks**Q8**
Binary Search
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack - Postfix Expression
5 marks

10/10 answered

Q7 Queue - Deque

5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {  
    if (front == 0) front = MAX - 1;  
    else front = front - 1;  
    deque[front] = data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int deque[MAX];  
6 int front = -1, rear = -1;  
7  
8 int isFull() {  
9     return ( (front == 0 && rear == MAX - 1) || (front == rear + 1) );  
10 }  
11  
12 int isEmpty() {  
13     return (front == -1);  
14 }  
15  
16 void insertFront(int data) {  
17     if (isFull()) {  
18         printf("Overflow: Deque is full. Cannot insert %d at front.\n",  
19             data);  
20         return;  
21     }  
22  
23     if (isEmpty()) {  
24         front = rear = 0;  
25     } else if (front == 0) {  
26         front = MAX - 1;  
27     } else {  
28         front = front - 1;  
29     }  
30  
31     deque[front] = data;  
32 }  
33  
34 void display() {  
35     if (isEmpty()) {  
36         printf("Deque: (empty)\n");  
37         return;  
38     }  
39  
40     printf("Deque: ");  
41     int i = front;  
42     while (i) {  
43         printf("%d ", deque[i]);  
44     }  
45 }
```

Test Input

stdin input (optional)

Output

```
Deque: (empty)  
Deque: 10  
Deque: 20 10  
Deque: 30 20 10  
Deque: 40 30 20 10  
Deque: 50 40 30 20 10
```

⚠ Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks

10/10 answered

Q8 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {  
    int mid = l + (r - l) / 2;  
    if (arr[mid] == x) return mid;  
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);  
    return binarySearch(arr, mid, r, x);  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

Test Input

stdin input (optional)

Output

```
Array: 1 3 5 7 9 11 13 15  
Search 7 -> Index 3  
Search 2 -> Index -1  
Search 15 -> Index 7
```

⌚ Pending manual review

```
1 #include <stdio.h>  
2  
3 int binarySearch(int arr[], int l, int r, int x) {  
4     if (l > r) return -1;  
5  
6     int mid = l + (r - l) / 2;  
7  
8     if (arr[mid] == x) return mid;  
9  
10    if (arr[mid] > x)  
11        return binarySearch(arr, l, mid - 1, x);  
12  
13    return binarySearch(arr, mid + 1, r, x);  
14 }  
15  
16 int main() {  
17     int arr[] = {1, 3, 5, 7, 9, 11, 13, 15};  
18     int n = sizeof(arr) / sizeof(arr[0]);  
19  
20     int x1 = 7;  
21     int x2 = 2;  
22     int x3 = 15;  
23  
24     printf("Array: ");  
25     for (int i = 0; i < n; i++) printf("%d ", arr[i]);  
26     printf("\n");  
27  
28     int result1 = binarySearch(arr, 0, n - 1, x1);  
29     printf("Search %d -> Index %d\n", x1, result1);  
30  
31     int result2 = binarySearch(arr, 0, n - 1, x2);  
32     printf("Search %d -> Index %d\n", x2, result2);  
33  
34     int result3 = binarySearch(arr, 0, n - 1, x3);  
35     printf("Search %d -> Index %d\n", x3, result3);  
36  
37     return 0;  
38 }
```

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Postfix Expression
5 marks

10/10 answered

Q9 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {  
    if (front == -1) return -1;  
    int data = queue[front];  
    front = (front + 1) % MAX;  
    return data;  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int queueArr[MAX];  
6 int front = -1;  
7 int rear = -1;  
8  
9 int isEmpty() {  
10     return (front == -1);  
11 }  
12  
13 int isFull() {  
14     return ((rear + 1) % MAX == front);  
15 }  
16  
17 void enqueue(int x) {  
18     if (isFull()) {  
19         printf("Enqueue(%d): Queue is Full\n", x);  
20         return;  
21     }  
22  
23     if (isEmpty()) {  
24         front = rear = 0;  
25     } else {  
26         rear = (rear + 1) % MAX;  
27     }  
28     queueArr[rear] = x;  
29 }  
30  
31 int dequeue() {  
32     if (front == -1) return -1;  
33  
34     int data = queueArr[front];  
35  
36     if (front == rear) {  
37         front = -1;  
38         rear = -1;  
39     } else {  
40         front = (front + 1) % MAX;  
41     }
```

Test Input

stdin input (optional)

Output

```
Test Case 1: Dequeue from empty  
queue  
front=-1 rear=-1  
dequeue() = -1  
  
Test Case 2: Enqueue 10,20,30
```

⌚ Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Postfix Expression
5 marks

10/10 answered

Q10 Stack - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {  
    if (!isdigit(expression[i])) push(expression[i] - '0');  
    else {  
        int op1 = pop();  
        int op2 = pop();  
        // perform operation: result = op1 + op2;  
        push(result);  
    }  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <ctype.h>  
3 #include <stdlib.h>  
4  
5 int stack[100];  
6 int top = -1;  
7  
8 void push(int val) {  
9     stack[++top] = val;  
10 }  
11  
12 int pop() {  
13     return stack[top--];  
14 }  
15  
16 int main() {  
17     char expression[] = "53+82-+";  
18     int i = 0;  
19  
20     while (expression[i] != '\0') {  
21         if (isdigit(expression[i])) {  
22             push(expression[i] - '0');  
23         } else {  
24             int op1 = pop();  
25             int op2 = pop();  
26             int result;  
27  
28             if (expression[i] == '+') result = op2 + op1;  
29             else if (expression[i] == '-') result = op2 - op1;  
30             else if (expression[i] == '*') result = op2 * op1;  
31             else if (expression[i] == '/') result = op2 / op1;  
32  
33             push(result);  
34         }  
35         i++;  
36     }  
37  
38     printf("%d\n", pop());
```

Test Input

stdin input (optional)

Output

48

Pending manual review